WAC 5.0: applications' traces, dated right

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WAC (Windows Artefacts Collector) is a free collector, in C++, that reads volumes raw and writes nothing on the examined machine, from XP SP1 to Windows 11 workstations and Server 2003 to 2025. Version 5.0 takes what applications keep outside the event logs, and dates each time written without a zone with the zone in force when it was written.

97 products, one table

Remote administration tools (TeamViewer, AnyDesk, RustDesk, ScreenConnect…), antivirus other than Defender, messaging, online storage, file transfer, VPN, virtualisation, server applications and logs (Netlogon, NPS, a certification authority), and what Windows keeps besides: 97 products in eleven groups, each with its folders, most from KAPE's targets, and its switch in `wac.yml`. PuTTY, WinSCP, WinRAR and 7-Zip are read in each user's hive. The folders are read raw, like the rest: nothing is opened by the system.

Copied, then converted

KAPE copies these files; WAC copies and converts them, without a decoder per product: a log becomes one element per line, with the time it starts with; XML, JSON and INI give their structure; a database or binary is listed and stays in the exhibit store. A value named like a password, a private key block of an OpenVPN profile, are never written: the JSON only says they exist. SSH private keys are not taken; crash dumps and the Office cache, which may hold documents, are off by default.

The time written without a zone

A clock time is not an instant. WAC converts it with the zone in force when it was written, drawn from the System log: every change of offset since Windows 10, the zone observed at boot and each day before. A change whose instant no event gives stays an interval, never an invented instant; a time lived twice gives its other possible instants. For each product present on the test machines, the assumed zone was measured against the clock: a provoked line, bracketed by the system time. TeamViewer, a line written under another zone, October's repeated hour: dated to the millisecond, as Windows converts them.

What the review found

The code and security review of 5.0 fixed two defects before release: a private key of a text file would have come out in clear in the JSON, and a forged NTFS index, naming a folder like its own pattern, looped the collection into a stack overflow. Each has its test, seen failing before the fix; 5.0's readers pass 46,000 mutated inputs with no read outside the buffer.

What remains to be measured

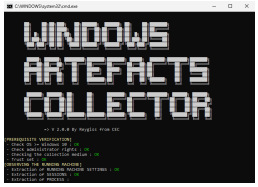
A witness proves that the table's path is taken, not that every version of a product writes there: most of the 97 products were not installed on the test machines, and their time is assumed local without verification. UltraViewer writes its date in the format of the account's culture: it is not read. The measured comparison with KAPE is still that of 29 October, before these targets.

In figures

Artefacts collected (4.0: 87)	99
Products in the traces' table	97
Checks of the cycle, Windows 11	677
Mutated inputs, 5.0's readers	46,000
Passwords or private keys written	0

Availability

Version 5.0.0, Windows XP SP1 to 11 and Server 2003 to 2025 (x86 and x64), GPL 3.0 licence: `https://github.com/dev-cec/WAC`. Manuals, validation, harmlessness and comparison files in French and English; cycles from 8 to 13 November 2026.



WAC collecting.